

7.2 Statistical techniques for Data Analysis (M2)

7.2.1 Headline information about the module

Module title	Statistical techniques for Data Analysis
Module NFQ level (only if an NFQ level can be demonstrated)	8
Module number/reference	M2
Parent programme(s) the plural arises if there are embedded programmes to be validated.	Higher Diploma in Science in Data Analytics for Business
Stage of parent programme	AWARD
Semester (semester1/semester2 if applicable)	1
Module credit units (FET/HET/ECTS)	ECTS
Module credit number of units	10
List the teaching and learning modes	Full time & Part time blended learning
Entry requirements (statement of knowledge, skill and competence)	See Section 4
Pre-requisite module titles	none
Co-requisite module titles	none
Is this a capstone module? (Yes or No)	No
Specification of the qualifications (academic, pedagogical and professional/occupational) and experience required of staff (staff includes workplace personnel who are responsible for learners such as apprentices, trainees and learners in clinical placements)	Academic and Professional: MSc required. However, NFQ Level 8 in Computer Science, Software Development, Software Engineering or equivalent is acceptable in cases where significant industrial experience is evident. Industry experience will be essential for those who do not have a postgraduate award. Software Development and Associated Experience: A minimum of 1-year experience required, 3 years of experience preferred. Pedagogical: Teaching experience is desired. Completion of postgraduate CPD/Certificate in Teaching and Learning or similar preferred. Experience in blended learning delivery required. In absence of experience, training will be provided.
Maximum number of learners per centre (or instance of the module)	120
Duration of the module	1 semester
Average (over the duration of the module) of the contact hours per week (see * below)	5 Hours

Module-specific physical resources and support required per centre (or instance of the module)						Physical resource requirements are 1 laptop or PC/workstation per student. On campus and online resources as per programme specification.				
Analysis of required learning effort										
*Effort while in contact with staff										
On campus Lecture / Classroom demonstrations		Mentoring and small-group tutoring		Other (Reflective development, directed reading/group work)		Online classes & Directed e-learning (hours)	Independent learning (hours)	Other hours (specify)	Work-based learning hours of learning effort	Total effort (hours)
Hours	Minimum ratio teacher/learner	Hours	Minimum ratio teacher/learner	Hours	Minimum ratio teacher/learner					
10	1:16	10	1:5	10	1:16	30	190	N/A	N/A	250
Allocation of marks (within the module)										
				Continuous assessment	Supervised project	Proctored practical examination	Proctored written examination	Total		
Percentage contribution				100%				100%		

7.2.2 Module aims and objectives

The aim of this module is to provide the learner with understanding of:

1. Numerical and statistical tools used to describe and summarise data.
2. The utility and application of inferential statistical methods.
3. The purpose and limitations of regression analysis and modelling.
4. The laws of probability and their application to data analysis.
5. Software tools used for the analysis of business data

7.2.3 Minimum intended module learning outcomes

On successful completion of this module the learner will be able to:

1. Explore and evaluate datasets using descriptive statistical analyses.
(Linked to PLO 1)
2. Formulate and test hypotheses within a business context using appropriate statistical techniques

and both evaluate and communicate the results effectively to peers and team members.

(Linked to PLO 3, PLO 6)

3. Apply regression analysis to appropriate datasets and demonstrate an awareness of the limitations of regression models.

(Linked to PLO 2)

4. Use and understand current software tools and languages to produce result sets from existing data(e.g. Excel, R, Python).

(Linked to PLO 4)

7.2.4 Rationale for inclusion of the module in the programme and its contribution to the overall IPLOs

Numerical methods, particularly those pertaining to statistics and probability, are central to the domain of data analytics. This module will equip the learner with the fundamental statistical skills that are immediately applicable to basic data analysis tasks as well as serving as a foundation for more sophisticated techniques introduced in later modules.

7.2.5 Information provided to learners about the module

A copy of the Module Descriptor will be provided to learners at the start of the module via the College LMS (Moodle)

7.2.6 Module content, organisation and structure

The following indicative syllabus contains a learner reflective component as outlined in the teaching and learning strategy for this programme.

Content
Descriptive Statistics • Measures of central tendency - mean, median, mode. • Measures of dispersion - range, variance, standard deviation. • Other measures - kurtosis and skewness

Inferential Statistics • Sampling • Hypothesis formulation and testing. • Alpha and p-values. • Parametric tests – e.g. Student’s t-test, F test, ANOVA • Non-parametric tests – e.g chi-square test • The Central Limit Theorem. • Confidence Intervals • Type I & II errors • Statistical power	
Probability • Laws of probability • Conditional and marginal probability • Bayes’ Theorem • Common probability distributions • Expected value	
Regression • Correlation • Simple Linear Regression • Multiple Linear Regression	

7.2.7 Module teaching and learning (including formative assessment) strategy

To provide the learner with a strong foundation in the core topics covered during the lectures, practical sessions will reinforce lecture content and provide supervised time to complete some assessment tasks. Sessions will be interactive, with instructor-led example exercises highlighting important topics discussed in lectures.

To provide formative assessment for this module the learner will:

- Be provided an opportunity at the beginning of each week to engage in group discussion on the material covered the previous week, thereby allowing reflection and ensuring their competency.
- Complete student-suggested tasks in a peer learning environment to encourage collaboration and allow learners to self-evaluate their current knowledge while gaining new knowledge and insights (this strategy links directly to PLO 7, PLO 8)
- Join additional discussions covering any lab-based exercises which have been provided to the learner.

7.2.8 Work-based learning and practice-placement

Not Applicable

7.2.9 E-learning

Collaborative blended learning strategies will be utilised for this module to ensure peer learning can be experienced not only through face-to-face traditional learning but also through online approaches. The online element will be achieved through a variety of interactive methods, including discussion forums, collaborative blogs and wikis, virtual labs and classrooms, group online supervision, interactive demonstrations and e-portfolios. This integrated learning approach ensures learning can be both reflective and collaborative while developing the efficacy of the individual student as the module progresses.

7.2.10 Module physical resource requirements

Physical resource requirements are 1 laptop or PC/workstation per student. On campus and online resources as per programme specification.

7.2.11 Reading lists and other information resources

7.2.12 Specifications for module staffing requirements

Academic and Professional: MSc required. However, NFQ Level 8 in Computer Science, Software Development, Software Engineering or equivalent is acceptable in cases where significant industrial experience is evident. Industry experience will be essential for those who do not have a postgraduate award.

Software Development and Associated Experience: A minimum of 1 year experience required, 3 years of experience preferred.

Pedagogical: Teaching experience is desired. Completion of postgraduate CPD/Certificate in Teaching and Learning or similar preferred. Experience in blended learning delivery required. In absence of experience, training will be provided.

7.2.13 Module summative assessment strategy

Mapping of summative and formative assessment: MIMLO to MIPLO

	Module Learning Outcomes				Programme Learning Outcomes								
	1	2	3	4	1	2	3	4	5	6	7	8	
Summative Assessment	✓	✓	✓	✓	✓	✓	✓	✓		✓	✓	✓	

7.2.14 Sample assessment materials

Due to the nature of the integrated assessment strategy approach for this programme, some integrated assessment materials will not be available until faculty have been assigned, at which point the time needed for the intense collaboration required will be set aside to create assessment based on this strategy. Examples will be provided to the panel.

